

IS 477 Interim Status Contribution Report - Siddhu Sami

For what we accomplished so far, I was responsible for creating a clean folder structure in google drive, cleaning the salaries dataset, merging the two datasets, and creating visualizations for the merged dataframe. I will outline my steps below.

Salary Data Processing:

- The salary data was filtered to only contain the relevant columns of "Player", "Salary", "Pos", "Year", "Team"
- A small but significant portion of players were listed as dual positions (e.g. SF-SG instead of SF). We made the decision to assign these players to the first position listed
- Then, salary counts were grouped by position and team and normalized, resulting in each team, each season, having a salary distribution by position to capture spending allocation
- This spending allocation data was then linked to the season/team results dataset, so we can eventually compare spending allocation to team success

Merged Data: We created a merged pandas dataframe of the salary data and the seasons data. This data frame includes every season from every NBA team that has 17 seasons in the dataset from 2005 onwards and includes the percentage of their total salary spent on each position. This dataframe will be used to perform analysis on NBA teams' salary allocation and seek to find relations between this and team success. Which can be determined by multiple different metrics including regular season wins, regular season offensive and defensive performance, overall player performance, and if the team won an NBA Championship.

Merged Dataframe Exploratory Data Analysis

- We made sure that there were no null values that may have popped up during the process of merging the two datasets.
- We noticed that there were some rows where one or more of the columns with percentage spent were 0. We decided not to remove these columns because this indicate a team decision to not spend on that position
- We also created visualizations using Matplotlib. The first one was created in order to see the average percentage of team salary allocated to each position over all seasons and every team. We also did the same thing for only teams that won the championship that season. Lastly we created a graph that shows the average all NBA teams' percentage of a team's salary that was allocated to each position over every season that was included in our dataset which was from 2005-2025



